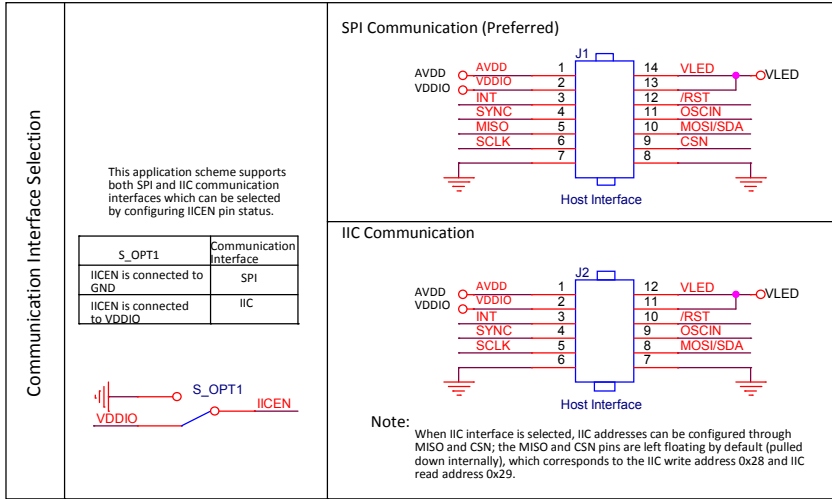


Host Interface



- Notes:**
1. SYNC and OSCIN are optional, depending on the specific application of customers.
 2. If no I/O pin is available on the host side for module /RST control, please make sure AVDD power supply (LDO) can be powered up or down through software; otherwise, the sensor may fail to be reset when malfunctions occur, and thus, it is recommended that /RST should be retained.
 3. The mainboard should supply power to VDDIO of the module, to ensure that the communication voltage level on the host side is identical with that on the module side; if the supply voltages of AVDD and VDDIO are identical, these two pins can be shorted together (share the same power supply).

System Power Supply Scheme

	VLED Power Supply	VDDIO Power Supply	Notes
Scheme	2.1-3.3V ≥20mA	1.8-3.3V ≥20mA	1. When the supply voltages of AVDD and VDDIO are identical, these two pins can be shorted together (share the same power supply), for example, 3.3V, ≥20mA. 2. Requirement on power supply noise: <100mVpp @0-1MHz bandwidth.

VLED Power Supply Scheme

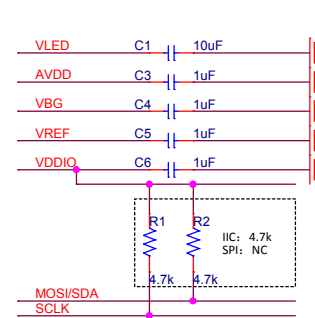
	VLED Power Supply	Remark:
Scheme	3.3V-5.3V ≥200mA	1. The relationship between LED voltage drop and VLED power supply are as follows: 25mA application scenario: LED voltage drop $V_f \leq$ VLED voltage-0.35V. 50mA application scenario: LED voltage drop $V_f \leq$ VLED voltage-0.40V. 100mA application scenario: LED voltage drop $V_f \leq$ VLED voltage-0.50V. 200mA application scenario: LED voltage drop $V_f \leq$ VLED voltage-0.70V. 2. 5.0V power supply is recommended. 3. If the LEDs are discrete components, the supply voltage of the IR LED can be 3.3V, and that of the Green/Red LED is recommended to be 5V, so as to reduce the power consumption of IR LED.

Note: If VLED is powered by the battery, evaluation on whether the voltage/current is able to meet the application requirements when the battery power is low should be implemented.

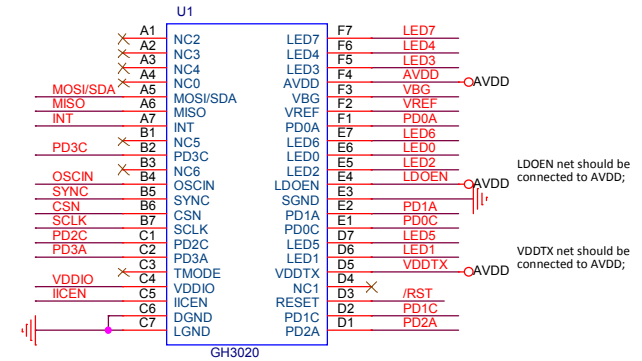
Disclaimer

Information contained in this document is intended for your convenience only and is subject to change without prior notice. It is your responsibility to ensure its application complies with technical specifications. Shenzhen Goodix Technology Co., Ltd. (hereafter referred to as "Goodix") makes no representation or guarantee for this information, express or implied, oral or written, statutory or otherwise, including but not limited to representation or guarantee for its application, quality, performance, merchantability or fitness for a particular purpose. Goodix shall assume no responsibility for this information and relevant consequences arising out of the use of such information. Under the protection of Goodix intellectual property rights, no license may be transferred implicitly or by any other means.

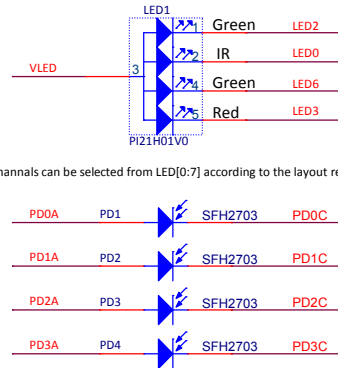
GH3020



- Notes:**
1. IIC pull-up resistors can be placed either on the mainboard or on FPC; please do not place them repeatedly.
 2. When AVDD and VDDIO are shorted together (share the same power supply), C6 should be left unconnected.



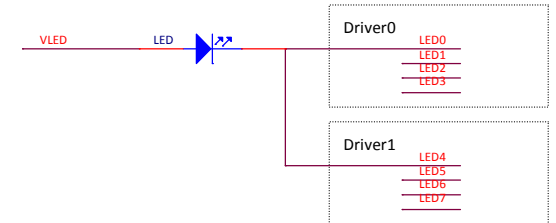
LED&PD



Note: LED channels can be selected from LED[0:7] according to the layout requirements.

Note: LED and PD should be selected according to the optical path designs of the specific products.

Optional



Note: If the dual-driver solution is required, select one channel from LED[0:3] and LED[4:7] respectively and connect them in parallel; the load driving capability of VLED should be no less than 400mA.